

Bridge

Infrastructure: Bridge

Budget: \$10,00,000

Location: Dubai, UAE

Target Year: 2030

Risk Level: LOW

Risk Score: 0.57/100

CLIMX Grounded Resilience Engine Report

Project Overview

The proposed bridge infrastructure in Dubai requires careful consideration of climate resilience strategies to mitigate the impacts of flooding. Given the recent severe flooding in April 2024 and projected sea level rise, it is essential to implement effective measures to ensure the bridge's stability and safety.

STRATEGY 1

Elevated Bridge Design

The bridge will be designed with elevated piers and decks to provide a minimum of 1.5 meters of freeboard above the projected flood levels. This will ensure that the bridge remains accessible and functional during flood events, reducing the risk of damage and disruption to traffic. The elevated design will also allow for the incorporation of flood-resistant materials and construction techniques.

STRATEGY 2

Flood-Resistant Materials and Construction

The bridge will be constructed using flood-resistant materials, such as stainless steel or fiber-reinforced polymer (FRP) rebar, to minimize the risk of corrosion and damage from floodwaters. The bridge's foundation and substructure will be designed to withstand the forces exerted by floodwaters, including scour and erosion. This will involve the use of specialized coatings and surface treatments to protect the materials from the corrosive effects of seawater.

STRATEGY 3

Integrated Drainage and Water Management System

The bridge will be designed with an integrated drainage and water management system, incorporating features such as culverts, catch basins, and stormwater management ponds. This system will be designed to capture and redirect floodwaters away from the bridge, reducing the risk of flooding and erosion. The system will also include green infrastructure elements, such as vegetated swales and rain gardens, to enhance stormwater management and mitigate the urban heat island effect.

RESILIENCE SCORE: 92%

